Electromagnetic Wave Optics

Part I: Mathematical Background and Basic Definitions

The following subjects will be covered

- Maxwell's Equations: Review
- The EM Wave Equation and Solutions.
- The Refractive Index Relation.
- Light Wave Power Density.

EM Wave Optics



EM Wave Optics



EM Wave Optics



EM Wave Optics





EM Wave Optics



Hale-Bopp Comet



Maxwell's Equations

Review of the fundamental laws of electricity and magnetism.

Fields	$\vec{E} = \frac{\vec{F}_e}{q}$	$(\vec{v} \times \vec{B}) = \frac{\vec{F}_m}{q}$
Flux densities	$\vec{D} = \varepsilon \vec{E}$	$\vec{B} = \mu \vec{H}$
Gauss's Law	$\vec{\nabla} \cdot \vec{D} = \rho_v$	$\oint \vec{D} \cdot \overrightarrow{ds} = Q_{enc}$
Gauss's Law	$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot \overrightarrow{ds} = 0$
Ampere's Law	$\vec{\nabla} \times \vec{H} = \vec{J}$	$\oint \vec{H} \cdot \overrightarrow{dl} = I_{enc}$
Faraday's Law	$\mathcal{E}mf = -\frac{\partial \varphi}{\partial t}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Charge conservation
The continuity equation

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$$

$$\vec{J} = \sigma \vec{E}$$

Basic Quantities

(V/m) $\vec{E}, \vec{D}$ (C/m^2)

(A/m) $\vec{H}, \vec{B}$ (Wb/m^2)

(A/m^2) $\vec{J}, \rho_v$ (C/m^3)

Material Properties

ε, μ, σ

(F/m) (H/m) (S/m)

Maxwell's Equations

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \qquad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{D} = \rho_v \qquad \nabla \cdot \vec{B} = 0$$

$$\vec{D} = \varepsilon \vec{E} \qquad \vec{B} = \mu \vec{H} \qquad \vec{J} = \sigma \vec{E}$$

Maxwell's Equations

Constitutive Relations

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\mu \frac{\partial (\vec{\nabla} \times \vec{H})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E} = -\mu \frac{\partial}{\partial t}(\vec{J} + \frac{\partial \vec{D}}{\partial t})$$

$$\vec{\nabla}^2 \vec{E} - \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E}$$

$$\vec{\nabla}^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

The EM Wave Equation

The EM Wave Equation

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial\vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Write the wave equation in this form

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \qquad \nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$$

Wave speed $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$ For free space $v = \dfrac{1}{\sqrt{\mu_o\varepsilon_o}} = 3.0\times10^8$ (m/s)

1) Electromagnetic waves exist.

2) The wave is composed of two interacting quantities E and H fields.

3) EM waves travel with the speed of light.

4) Light is an EM waves.

TH Plane-wave Solution in Lossless Media

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Time-harmonic waves are generated by time-harmonic sources.

$\vec{E}(r,t) = \vec{E}(r)\,e^{j\omega t}$ r here represents the space variable. $e^{j\omega t}$ Sinusoidal, single frequency

$$\vec{\nabla}^2\vec{E}(r) + k^2\vec{E}(r) = 0$$
$$\vec{\nabla}^2\vec{H}(r) + k^2\vec{H}(r) = 0$$

Helmholtz wave equation

$$k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{v}$$

Wave number (rad/m) (Real number)

Refractive index

$$n = \sqrt{\mu_r\varepsilon_r} \qquad v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \qquad k = \frac{n\omega}{c}$$

The Uniform Plane-wave Solution

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

TH Wave equation $\nabla^2\vec{E}(x,y,z) + k^2\vec{E}(x,y,z) = 0$ ➡ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)\vec{E}(x,y,z) = 0$

Consider a one-dimensional case (z) with one field component only.

$$\frac{\partial^2 E_x}{\partial z^2} + k^2 E_x = 0$$

$$E_x(z) = A\,e^{-jkz} + B\,e^{jkz}$$

Plane wave Solution

A and B are constants.

Put back the time-harmonic factor

$$E_x(z,t) = A\,e^{j(\omega t - kz)} + B\,e^{j(\omega t + kz)}$$

Wave travelling along the $+ve$ z direction

Wave travelling along the $-ve$ z direction

Take the $+z$ wave

$$E_x(z,t) = E_o\,e^{j(\omega t - kz)}$$

$$E_x(z,t) = E_o\cos(\omega t - kz) + j\,E_o\sin(\omega t - kz)$$

Solution Solution

Take the sine wave solution

$$E_x(z,t) = E_o\sin(\omega t - kz)$$

$$\vec{E}(z,t) = E_o\sin(\omega t - kz)\;\vec{a}_x$$

Same definitions for E_o, ω, k

Time-harmonic (Sinusoidal) Trans Waves

Electric field Part of the wave

Amplitude

Phase

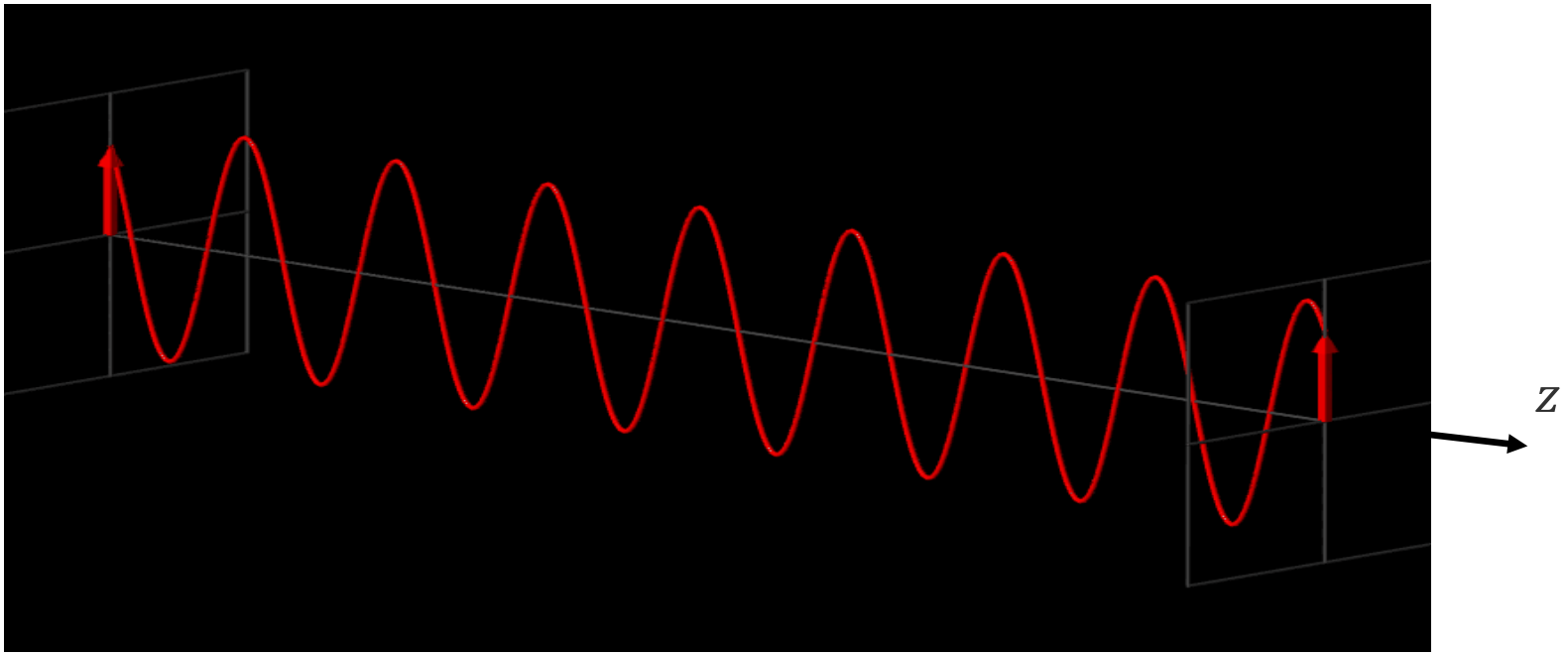
$$\vec{E}(z,t) = E_o \sin(\omega t \pm kz)\ \vec{a}_x$$

Field orientation/polarization

$\omega = 2\pi f$ Frequency

$k = \frac{\omega}{v}$ Wave number

Direction of wave propagation



The Uniform Plane-wave Solution

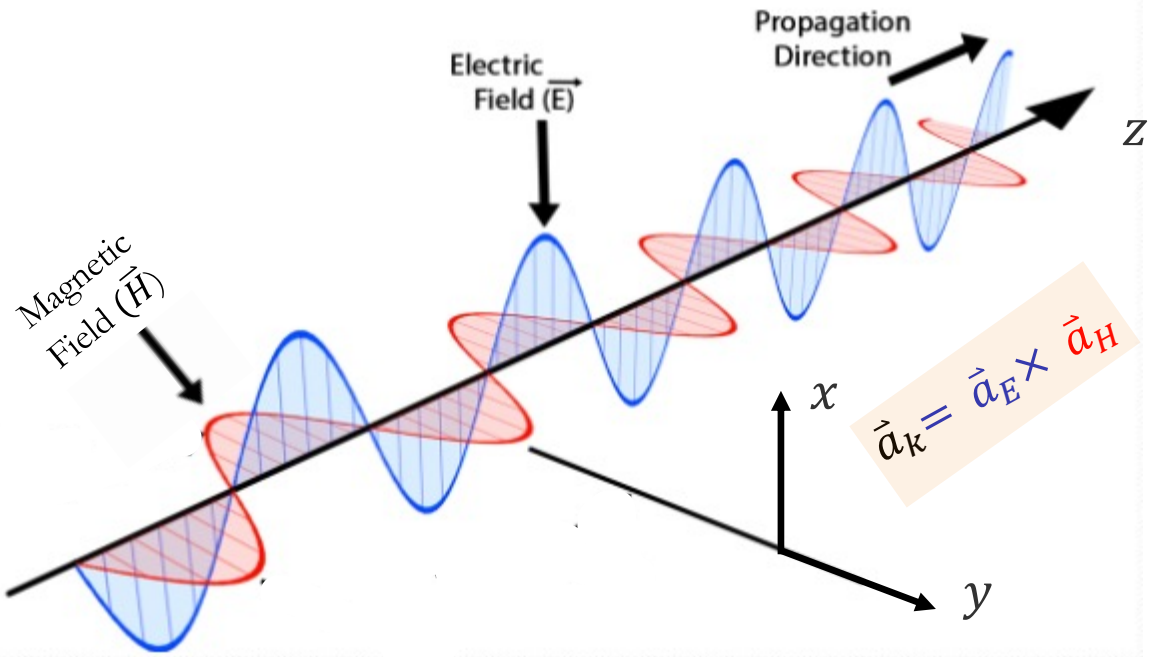
$$\vec{E}(z,t) = E_o \sin(\omega t - kz)\ \vec{a}_x$$

What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$$\vec{E}(z) = E_o\, e^{-jkz}\ \vec{a}_x$$

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$



For the 1D TH case, we have

$$\vec{H} = \frac{1}{-j\omega\mu}\ \vec{\nabla} \times \vec{E}\ = \frac{1}{-j\omega\mu}\ (-jk) E_o\, e^{-jkz}\ \vec{a}_y\ = \frac{k}{\omega\mu}\ E_o\, e^{-jkz}\ \vec{a}_y \qquad k^2 = \omega^2 \mu\varepsilon$$

$$\vec{H} = \sqrt{\frac{\varepsilon}{\mu}}\, E_o\, e^{-jkz}\, \vec{a}_y$$

$\sqrt{\frac{\mu}{\varepsilon}}$ has units of impedance (Ω)

$$\eta = \frac{E_o}{H_o} = \sqrt{\frac{\mu}{\varepsilon}}$$ Plane wave impedance

$$H_o = \frac{E_o}{\eta}$$

$$\vec{H} = H_o\, e^{-jkz}\ \vec{a}_y$$

For free-space $\eta_o = \sqrt{\frac{\mu_o}{\varepsilon_o}} = 377\ \Omega$

Other media $\eta = \frac{\eta_o}{n}$

TH Plane-wave Solution in Lossy Media

Complex Refractive index

$$n = \sqrt{\mu_r \varepsilon_r} = n' - jn''$$

Complex propagation constant

$$k = \frac{n\omega}{c}$$

$$k = \frac{\omega}{c} n'' + j \frac{\omega}{c} n'$$

$$k = \alpha + jk$$

Field attenuation

Phase constant

$$\alpha = \frac{\omega}{c} n'' \qquad k = \frac{\omega}{c} n'$$

Refractive index

$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

Extinction coefficient

$$n'' = -\frac{\varepsilon_r''}{2n'}$$

Other relations

$$\varepsilon_r' = (n')^2 + (n'')^2$$

$$\varepsilon_r'' = -2n'\, n''$$

Wave speed

$$v = \frac{c}{n'}$$

Wave impedance

$$\eta = \frac{\eta_o}{n} = \frac{\eta_o}{n' - jn''}$$

The Plane-wave Solution in Lossy Media

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

Consider a one-dimensional case (z) with one component only.

$$\frac{\partial^2 E_x}{\partial z^2} - k^2 E_x = 0 \qquad k \text{ is complex}$$

$$E_x(z) = A\,e^{-kz} + B\,e^{kz}$$

Wave travelling along the $+ve\ z$ direction

Wave travelling along the $-ve\ z$ direction

Complex wave impedance $\eta = \frac{E_o}{H_o} = \frac{j\omega\mu}{\gamma} = |\eta|\,\angle\theta_\eta$

$$\vec{a}_k = \vec{a}_E \times \vec{a}_H$$

Take the $+z$ wave

$$E_x(z) = E_o\,e^{-(\alpha+jk)z}$$

$$E_x(z) = E_o\,e^{-\alpha z}\,e^{-jkz}$$

Attenuation Phase

Take the sine wave solution

$$E_x(z,t) = E_o \sin(\omega t - kz)\,e^{-\alpha z}$$

$$\vec{E}(z,t) = E_o \sin(\omega t - kz)\,e^{-\alpha z}\,\vec{a}_x$$

$$\vec{H}(z,t) = H_o \sin(\omega t - kz)\,e^{-\alpha z}\,\vec{a}_y$$

Power Density in a Plane Wave

Consider the following Plane wave represented by E and H.

$$\vec{E} = E_o \cos(\omega t - kz)\, e^{-\alpha z}\ \vec{a}_x$$

$$\vec{H} = H_o \cos(\omega t - kz - \theta_\eta)\, e^{-\alpha z}\ \vec{a}_y$$

$$\vec{S} = \vec{E} \times \vec{H}$$

Poynting vector (Wave power density W/m^2)

$$\vec{S} = E_o H_o \cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

$$\vec{S} = \frac{{E_o}^2}{|\eta|}\cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

The time-average power density is given by

$$\vec{S}_{ave} = \frac{1}{T}\int_0^T \vec{S}\ dt\ \vec{a}_z$$

(W/m^2)

For a plane wave,

$$\vec{S}_{ave} = \frac{1}{2}\, Re\left[\vec{E} \times \vec{H}^*\right] \Rightarrow \vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\ e^{-2\alpha z}\ \vec{a}_z$$

Total average power within some area

$$P_{ave} = \int_A \vec{S} \cdot \overrightarrow{dA} \quad \text{(W)}$$

Example

Consider an EM wave given by $\vec{E} = 4\ \sin(8.8\pi\times10^{14}t - kx)\ \vec{a}_z$ (V/m) travelling in a non-magnetic medium of n =3.8. Find k, η and $\vec{S}_{ave}$.

Lossless medium $\alpha = 0$

Frequency $\omega = 8.8\pi\times10^{14}$ rad/s

Refractive index $n = 3.8$ (real)

$$k = \frac{\omega}{c}n = \frac{8.8\pi\times10^{14}}{3\times10^{8}}\ (3.8) = 3.5\times10^{7}\ \text{rad/m}$$

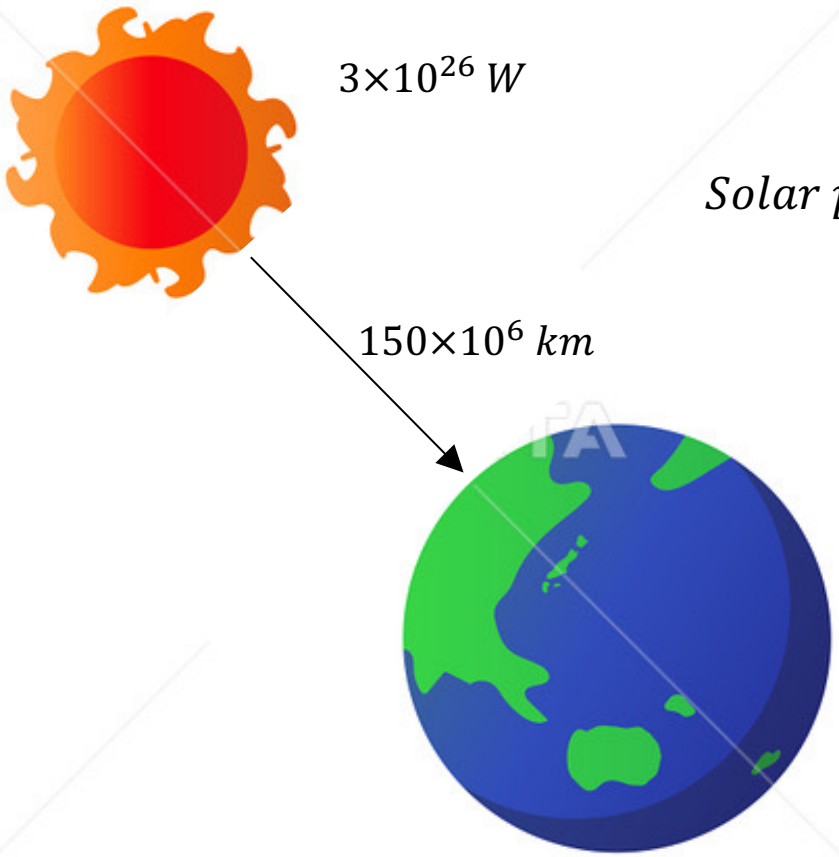
$$v = \frac{c}{n} = \frac{3\times10^{8}}{3.8} = 0.79\times10^{8}\ \text{m/s}$$

$$\eta = \frac{\eta_o}{n} = \frac{377}{3.8} = 98.6\ \Omega$$

$$\vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\ e^{-2\alpha z}\ \vec{a}_x$$

$$\vec{S}_{ave} = \frac{4^2}{2(98.6)}\ \vec{a}_x = 81.1\ \vec{a}_x\ \text{mW/m}^2$$

Example



$$Total\ solar\ power = 3\times10^{26}\ W$$

$$Solar\ power\ density\ at\ Earch\ surface = \frac{3\times10^{26}}{4\pi r^2}$$

$$S_{ave} = \frac{{E_o}^2}{2|\eta|} \approx 1\ kW/m^2$$

$$E_o = 890\ V/m$$

EM Wave Optics

EM wave optics enables us to study important light behavior, such as

- Reflection and transmission of light.
- Polarization of light.
- Light-matter interaction.
- Absorption of light.
- Scattering of light.

And many more!

End of Part I

Q & A

Send me an email if you have any questions.